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Olist E-Commerce Fulfillment Analysis

Does delivery timing relative to estimated dates drive measurable differences in customer review scores, and can that timing be predicted before it happens?

Dataset	Olist Brazilian E-Commerce · Kaggle CC0
Scope	09/2016 – 08/2018 · 96,470 delivered orders
Tools	SQL · Python · pandas · scipy · scikit-learn · matplotlib
Date	August – September 2026

Executive Summary

Testing reveals a very strong association ($p\text{-value} < 0.001$) between delivery timing relative to the estimated delivery date and review scoring.

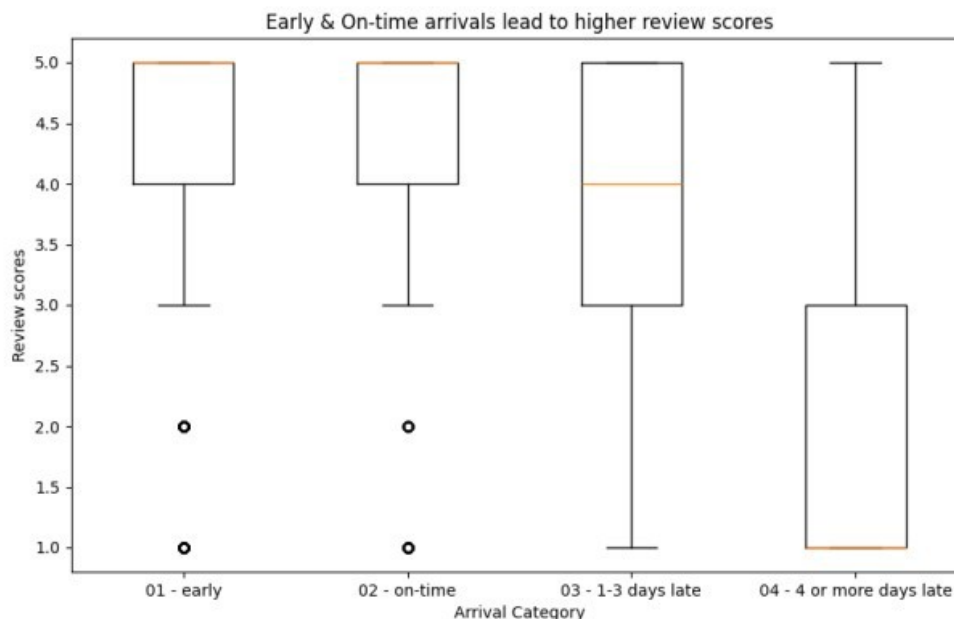
A subsequent modeling phase attempted to translate this association into an operational prediction model. Despite multiple tuning strategies, no combination of available features could reliably flag at-risk orders ahead of delivery, a finding detailed in Phase 2: Predictive Modeling.

Dataset & Scope

Data derived from the Olist e-commerce Public Dataset, published on Kaggle. The dataset ranges from 09/2016–08/2018. The scope of this analysis only covers orders with an order status of "delivered". After filtering on delivered, there were approximately 3,000 entries that had postal code mismatches that were omitted. This represents about 2% of the remaining 96,470 entries, that were marked "delivered".

Phase 1: Analytical Findings

The platform is doing well with on-time deliveries at 92.08%. Focusing on the opportunity area of late deliveries, when a delivery arrives 4 days or more later than expected, the customer review score plummets, on average to a 1.93 rating. By contrast, the early deliveries average a 4.3 rating.

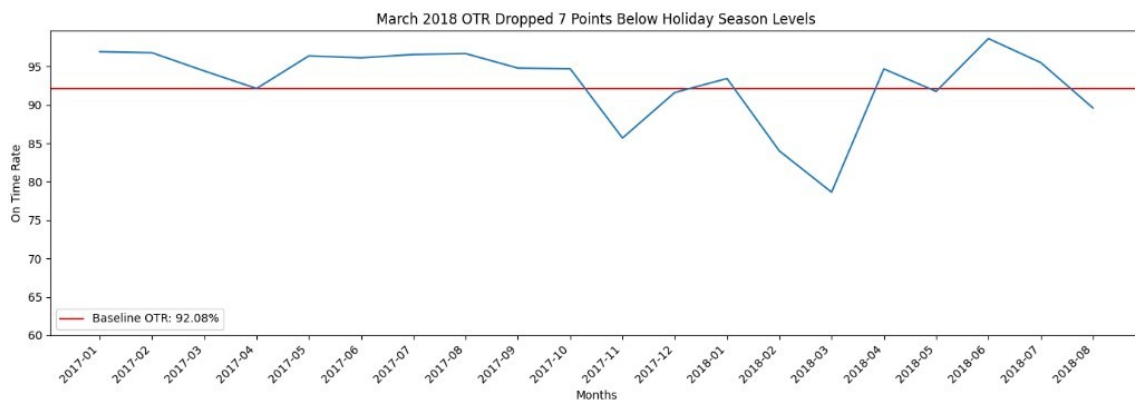


In an effort to determine what variables show a strong connection to late arrivals, a Pearson correlation test was done between the seller-customer distance and the corresponding review scores. The score came back at -0.077 , revealing there was effectively no correlation between these two variables.

Although it is clear that distance and delay connection is non-existent, we cannot overlook the fact that each state has its own OTR ranking. Following that thread, the state of MA did reveal itself as a bottom performer in the OTR category, by state, at 77%, this is notably 15 points below the platform average. With the understanding that distance is not a contributing factor in delays, the next question focused on shipping cost as a percentage of the weight.

Tests on shipping cost relative to freight weight was done to determine any correlations there. Again, testing revealed there was no meaningful correlation. Each seller was ranked based upon a weighted composite score that segmented them based upon volume, on-time rate and review score. The composite was designed to balance these three measures and reveal a clear value ranking of each seller. Sellers were ranked into low, mid and high tiers. The composite was crafted by indexing the three measures & weighing them as: volume (0.45), OTR (0.40) & review score (0.15). Having analyzed down to the seller level, the analysis then turned to the delivery time variance over time.

Analysis shows holiday peak season OTR dips to 88.47%. This marks an almost 4 point drop in this two month peak season. The analysis reveals a more concerning anomaly, March 2018, with its 78.6% OTR. Making for a 13 point drop from platform average and almost 10 points worse than peak holiday season. The data is silent on the reasoning for the collapse, more study and a wider scope are needed to discern. Data collection ended in Aug 2018, the partial month data has been excluded from the dataset, so as not to misrepresent the month. With the analysis done on the seasonal OTR, the study looked to determine if there were any connections between the expected and actual arrival times and review scores.



Kruskal-Wallis test revealed a p-value < .001 and a test statistic of 10,089. These results confirm the association of arrival times and the review scores. Arrival times were bucketed into 4 ranges (early, on-time, 1-3 days late, 4+ days late). A Mann-Whitney U test similarly revealed a p-value < .001. This further confirmed the association between the expected arrival time (early vs. not early) and the resulting review scores.

Phase 2: Predictive Modeling

Phase 1 established a strong statistical association between delivery timing and review scores. Phase 2 tested whether that association could be operationalized: could a model, using information available at or before the time an order is placed, flag orders at risk of arriving late in time for an operations team to intervene?

Q9 — Classification: A logistic regression classifier was built on a feature mart of order, seller, and timing attributes to predict late delivery. The default model did not produce an operationally productive classifier. Applying `class_weight='balanced'` raised recall from 0.2% to 12%, at the cost of precision dropping from 0.22 to 0.06 and accuracy falling 10 points. Lowering the decision threshold from 0.5 to 0.3 raised recall further to 31%, but accuracy fell to 62% with precision dropping slightly to 0.05. This result mirrors Phase 1's distance-delay finding: the available data does not capture the operational factors actually driving late delivery.

Q10 — Feature Importance: Standardized coefficients showed minimal predictive weight overall, consistent with the near-zero distance correlation and the classifier's inability to separate late orders in Q9. March had one of the highest predictive weights for late arrival, corroborating the Phase 1 OTR-by-month finding. São Paulo was a strong predictor of on-time arrival, plausibly reflecting the state's developed logistics network. December was checked directly and confirmed negligible (`std_coef` 0.038 vs. March's 0.357).

Q11 — Capture Rate vs. Random Selection: Ranking orders by predicted probability and comparing the top 5%, 10%, and 20% against random baselines produced capture rates of 5.73%, 10.36%, and 18.21%, against random baselines of 5%, 10%, and 20% respectively. Resulting lift values (1.15, 1.04, 0.91) sit at or below 1.0 across the board, meaning the model performs no better, and at wider capture widths slightly worse, than random selection.

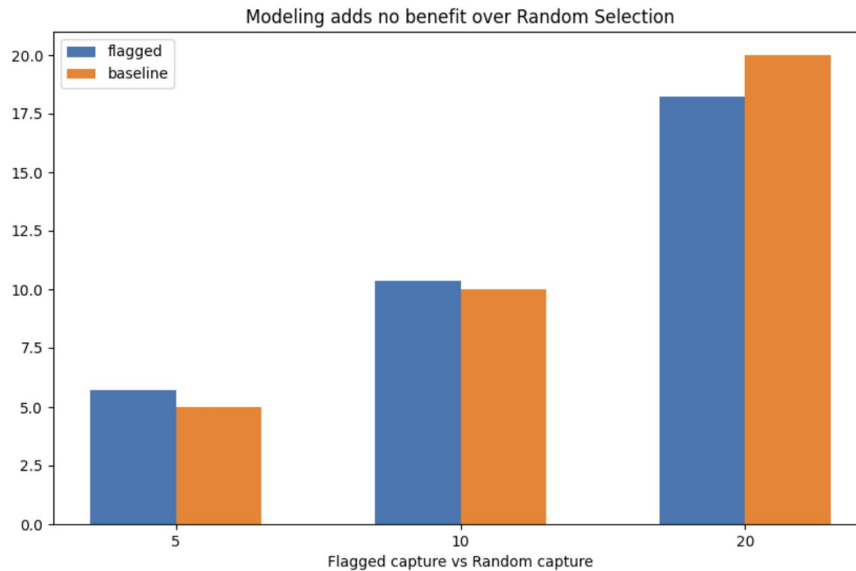


Figure: Capture rate at top 5/10/20% vs. random baseline. Bars track their baselines almost exactly, the signature of a model with no meaningful lift.

Q12 — Capacity-Constrained Precision & Recall: Assuming an intervention team capacity of 5% of daily order volume (an estimate, since no actual team size exists to reference), the corresponding probability threshold was 0.575. At that cutoff, precision was 6.1% (roughly 94% of flagged orders would be false alarms), and recall was 5.73%, in line with random selection. Intervening at this threshold is not operationally feasible: the cost of acting on false alarms is not offset by any meaningful improvement in catching orders that will actually arrive late.

Taken together, Q9 through Q12 point to the same conclusion through four independent tests: classification, feature weight, ranked capture rate, and capacity-constrained precision/recall. The dataset available at order-placement time does not support a credible, operationally useful prediction of late delivery.

Limitations

E-commerce sales data is limited to Brazil, only, during the period between 09/2016–08/2018. These findings are only representative of this scoping. No conclusions are to be drawn about other regions or other time frames.

The March 2018 anomaly of the 15 point collapse is only revealed through the data. There is no supporting evidence to reveal the source of the collapse.

With respect to the delivery delays, the data does not reveal any reasoning. There is no way to determine if the delivery left the seller late, got lost, or any other possibility. The only definite is the product arrived after the expected date.

Although the analysis reveals a strong association between delivery timing and review scores, the data does not establish a causal relationship between the two.

The Phase 2 classifier's failure to discriminate late orders does not rule out that late delivery is predictable in principle; it indicates that the features available in this dataset at order-placement time are insufficient. Real-time signals such as carrier scan events or live traffic conditions were not available.

The 5% intervention capacity used in Q12 is an estimate, not a figure derived from an actual operations team's throughput; the conclusion holds directionally even if the true capacity differs, since precision and recall were both near baseline across the 5/10/20% range tested in Q11.

Appendix

Figure A: Seller-Customer Distance vs. Delivery Time Difference (Pearson coef: -0.077)

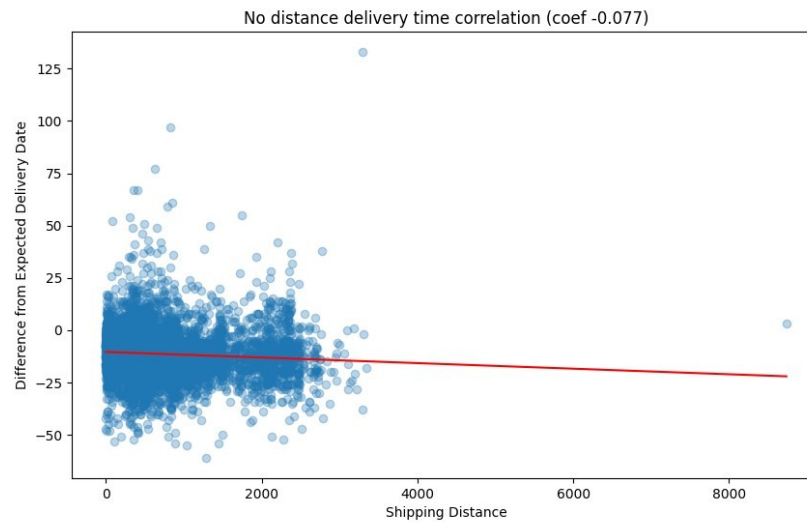


Figure B: Early vs. Not Early Arrival Review Score Distributions (Mann-Whitney U, $p < .001$)

